

$$= -\frac{1}{3} \begin{pmatrix} 2e^{2t} & e^{2t} \\ e^{st} & -e^{st} \end{pmatrix}$$

$$X_p = \begin{pmatrix} e^{-2t} & e^{-st} \\ e^{-2t} & -2e^{-st} \end{pmatrix} \int \frac{1}{3} \begin{pmatrix} 2e^{2t} & e^{2t} \\ e^{st} & -e^{st} \end{pmatrix} \begin{pmatrix} 3t \\ e^{-t} \end{pmatrix} dt$$

$$= \frac{1}{3} \begin{bmatrix} e^{-2t} & e^{-st} \\ e^{-2t} & -2e^{-st} \end{bmatrix} \int \begin{pmatrix} 6te^{2t} & te^{2t} \\ 3te^{st} & -e^{4t} \end{pmatrix} dt$$

$$= \frac{1}{3} \begin{pmatrix} e^{-2t} & e^{-st} \\ e^{-2t} & -2e^{-st} \end{pmatrix} \begin{pmatrix} 3te^{2t} - \frac{3}{2}e^{2t} + e^t \\ \frac{3}{5}te^{st} - \frac{3}{25}e^{st} - \frac{1}{4}e^{4t} \end{pmatrix}$$

$$= \frac{1}{3} \begin{pmatrix} 3t - \frac{3}{2} + e^{-t} + \frac{3}{5}t - \frac{3}{25} - \frac{1}{4}e^{-t} \\ 3t - \frac{3}{2} + e^{-t} - \frac{6}{5}t + \frac{6}{25} + \frac{1}{2}e^{-t} \end{pmatrix}$$

$$= \frac{1}{3} \begin{pmatrix} \frac{18}{5}t - \frac{81}{50} + \frac{3}{4}e^{-t} \\ \frac{9}{5}t - \frac{63}{50} + \frac{3}{2}e^{-t} \end{pmatrix}$$

$$X_p = \begin{pmatrix} \frac{6}{5}t - \frac{27}{50} + \frac{1}{4}e^{-t} \\ \frac{3}{5}t - \frac{21}{50} + \frac{1}{2}e^{-t} \end{pmatrix}$$

Solution General

$$X(t) = \begin{bmatrix} e^{-2t} & e^{-5t} \\ e^{-2t} & -2e^{-5t} \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} \frac{6}{5}t - \frac{27}{50} + \frac{1}{4}e^{-t} \\ \frac{3}{5}t - \frac{27}{50} + \frac{1}{2}e^{-t} \end{bmatrix}$$

PVI $X(0) = Y(0) = 0$

$$X(0) = \begin{bmatrix} a & b \\ a & -2b \end{bmatrix} = \begin{bmatrix} 27/50 - 1/4 \\ 27/50 - 1/2 \end{bmatrix}$$

$$\begin{bmatrix} a + b \\ a - 2b \end{bmatrix} = \begin{bmatrix} 29/100 \\ -2/25 \end{bmatrix}$$

$$\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1/6 \\ 37/300 \end{bmatrix}$$

$$X(t) = \begin{bmatrix} e^{-2t} & e^{-5t} \\ e^{-2t} & -2e^{-5t} \end{bmatrix} \begin{bmatrix} 1/6 \\ 37/300 \end{bmatrix}$$

$$d) \begin{cases} x' = x + 8y + e^{-t} \\ y' = x - y + te^{-t} \end{cases} \quad x(0) = y(0) = 0$$

$$\begin{bmatrix} 1 & 8 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} e^{-t} \\ te^{-t} \end{bmatrix}$$

$$\begin{bmatrix} 1-\lambda & 8 \\ 1 & -1-\lambda \end{bmatrix} = \begin{cases} (1-\lambda)(-1-\lambda) - 8 = 0 \\ -1 - \lambda + 1 + \lambda^2 - 8 = 0 \end{cases}$$

$$\lambda^2 - 9 = 0$$

$$\lambda = 3$$

$$\lambda = -3$$

Γ_{-3}

$$\begin{bmatrix} -1-3 & -8 \\ -1 & 1-3 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -4 & -8 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$V_1 = \begin{bmatrix} -2v \\ v \end{bmatrix} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$u + 2v = 0$$

$$v = v$$

$$u = -2v$$

E₃

$$\begin{bmatrix} -1 & 3 & -8 \\ -1 & 1 & 3 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$u - 4v = 0 \quad v = v$$

$$u = 4v$$

$$\begin{bmatrix} 3 & -8 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$V_2 = \begin{bmatrix} 4v \\ v \end{bmatrix} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$$

$$\begin{bmatrix} 1 & -4 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Soluciones rectas

$$x_1 = e^{-3t} \begin{pmatrix} -2 \\ 1 \end{pmatrix} \quad x_2 = e^{3t} \begin{pmatrix} 4 \\ 1 \end{pmatrix}$$

Matriz fundamental

$$\Phi = \begin{bmatrix} -2e^{-3t} & 4e^{3t} \\ e^{-3t} & e^{3t} \end{bmatrix}$$

$$\det \begin{pmatrix} -2e^{-3t} & 4e^{3t} \\ e^{-3t} & e^{3t} \end{pmatrix} = (-2e^{-3t})(e^{3t}) - (e^{-3t})(4e^{3t})$$

$$\Phi^{-1} = -\frac{1}{6} \begin{pmatrix} e^{3t} & -4e^{3t} \\ -e^{-3t} & -2e^{-3t} \end{pmatrix} \quad -2 - 4 = -6$$

$$x_p = \begin{pmatrix} -2e^{-3t} & 4e^{3t} \\ e^{-3t} & e^{3t} \end{pmatrix} \int \frac{-1}{6} \begin{pmatrix} e^{3t} & -4e^{3t} \\ -e^{-3t} & -2e^{-3t} \end{pmatrix} \begin{pmatrix} e^{-t} \\ te^{-t} \end{pmatrix} dt$$

$$= -\frac{1}{6} \begin{pmatrix} -2e^{-3t} & 4e^{3t} \\ e^{-3t} & e^{3t} \end{pmatrix} \int \begin{pmatrix} e^{2t} & -4te^{2t} \\ -e^{-4t} & -2te^{-4t} \end{pmatrix} dt$$

$$= -\frac{1}{6} \begin{pmatrix} -2e^{-3t} & 4e^{3t} \\ e^{-3t} & e^{3t} \end{pmatrix} \begin{pmatrix} \frac{e^{2t}}{2} - 2te^{2t} + e^{2t} \\ \frac{1}{4}e^{-4t} + \frac{1}{2}te^{-4t} + \frac{1}{8}e^{-4t} \end{pmatrix}$$

$$= -\frac{1}{6} \begin{pmatrix} 0 & (4t+1)e^{-t} & -\frac{3}{2}e^t \\ \frac{3}{4}e^{-t} & \frac{(-4t+1)}{2}e^t & \frac{9}{8}e^t \end{pmatrix}$$

$$X_p = \begin{bmatrix} 0 & \frac{(-4t-2)}{6}e^{-t} & \frac{1}{4}e^{-t} \\ -\frac{1}{8}e^{-t} & \frac{(4t-1)}{12}e^{-t} & -\frac{3}{16}e^{-t} \end{bmatrix}$$

$$e) \begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} -x_1 \\ x_2 \end{pmatrix} + \sec(t) \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad x_1(0) = 1 \\ x_2(0) = 0$$

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} \sec(t) \\ 0 \end{pmatrix} = \dot{x}$$

$$\begin{pmatrix} 0-\lambda & -1 \\ 1 & 0-\lambda \end{pmatrix} =$$

$$(-\lambda)(-\lambda) - (-1) = 0$$

$$\lambda^2 + 1 = 0$$

$$\lambda = \pm \sqrt{-1}$$

$$\lambda = i$$

$$\lambda = -i$$

→

$$\lambda = i$$

E-i

$$\begin{pmatrix} -(-i) & -1 \\ 1 & -(-i) \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} i & -1 \\ 1 & i \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$v_1 = \begin{pmatrix} 1 \\ i \end{pmatrix}$$

$$iu - v = 0$$

$$iu = v$$

$$u - iv = 0$$

$$u - u = 1$$

$$v = i$$

$$v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + i \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Sol: Real Part

$$x_1 = e^{-it} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + i \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right]$$

$$e^{-it} = [\cos(-t) + i \sin(-t)] \\ = [\cos(t) - i \sin(t)]$$

$$x_1 = [\cos(t) - i \sin(t)] \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + i \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right]$$

$$= \cos(t) \begin{pmatrix} 1 \\ 0 \end{pmatrix} + i \cos(t) \begin{pmatrix} 0 \\ 1 \end{pmatrix} - i \sin(t) \begin{pmatrix} 1 \\ 0 \end{pmatrix} - \sin(t) \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$= \cos(t) \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \sin(t) \begin{pmatrix} 0 \\ 1 \end{pmatrix} + i [\cos(t) \begin{pmatrix} 0 \\ 1 \end{pmatrix} - \sin(t) \begin{pmatrix} 1 \\ 0 \end{pmatrix}]$$

$$= \underbrace{\begin{bmatrix} \cos(t) \\ \sin(t) \end{bmatrix}}_{\text{IR}} + i \underbrace{\begin{bmatrix} -\sin(t) \\ \cos(t) \end{bmatrix}}_{\text{Imaginary}}$$

Generalizando

$$A = \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \rightarrow \Phi = e^{at} R(bt)$$

$$e^{At} = e^{at} R[bt] = e^{at} \begin{bmatrix} \cos bt & -\sin bt \\ \sin bt & \cos bt \end{bmatrix}$$

$$\Phi(t) \rightarrow a=0, b=1$$

$$\Phi(t) = e^{At} = \begin{bmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

$$x_p = \Phi(t) \int \Phi^{-1}(t) b(t) dt$$

$$\Phi^{-1}(t) = \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix} \quad \begin{array}{l} \cos^2 t + \sin^2 t = \det \\ 1 = \det \end{array}$$

$$\int \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix} \begin{pmatrix} \sec t \\ 0 \end{pmatrix} dt$$

$$\int \begin{pmatrix} \cos t \cdot \sec t \\ -\sin t \sec t \end{pmatrix} dt$$

$$\cos t \cdot \frac{1}{\cos t} = \int 1 = t$$

$$-\sin t \cdot \frac{1}{\cos t} = -\frac{\sin t}{\cos t}$$

$$-\int \frac{\sin t}{\cos t} dt$$

$$u = \cos t$$

$$du = -\sin t dt$$

$$dt = \frac{-1}{\sin t} du$$

$$\int \frac{1}{u} du = \ln(|\cos t|)$$

$$x = \begin{bmatrix} \text{cost} & \text{sent} \\ \text{sent} & \text{cost} \end{bmatrix} \begin{bmatrix} t \\ \ln(\text{cost}) \end{bmatrix}$$

$$\begin{bmatrix} t \text{ cost} + \text{sent} \ln(\text{cost}) \\ -t \text{ sent} + \text{cost} \ln(\text{cost}) \end{bmatrix}$$

$$x(t) = \begin{bmatrix} \text{cost} & -\text{sent} \\ \text{sent} & \text{cost} \end{bmatrix} + \begin{bmatrix} t \text{ cost} + \text{sent} \ln(\text{cost}) \\ -t \text{ sent} + \text{cost} \ln(\text{cost}) \end{bmatrix}$$

$$x(t) = \begin{bmatrix} \text{cost} & -\text{sent} \\ \text{sent} & \text{cost} \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} t \text{ cost} + \text{sent} \ln(\text{cost}) \\ -t \text{ sent} + \text{cost} \ln(\text{cost}) \end{bmatrix}$$

$$\begin{bmatrix} \text{cost} & -\text{sent} \\ \text{sent} & \text{cost} \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} t \text{ cost} + \text{sent} \ln(\text{cost}) \\ -t \text{ sent} + \text{cost} \ln(\text{cost}) \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$